

LEVEL 3: Array of Books

1. Objective

Store multiple book records instead of one book. Students learn an array of structures, indexing, traversal and basic complexity.

2. Data Structure / Algorithm Used

Array of Structures

3. Problem Statement

Extend the Library Management System by implementing the array of books module. Complete this module independently before integrating it with previous levels.

4. Algorithm / Working

Read $n \rightarrow$ store each Book in $\text{books}[i] \rightarrow$ traverse the array $\rightarrow$ display every record.

5. C Program

```
#include <stdio.h>
#define MAX_BOOKS 100
typedef struct {
    int id; char title[50]; char author[50]; int year; int available;
} Book;

int main() {
    Book books[MAX_BOOKS];
    int n,i;
    printf("Number of books: "); scanf("%d",&n);
    if(n<1 || n>MAX_BOOKS) { printf("Invalid number\n"); return 0; }

    for(i=0;i<n;i++) {
        printf("\nBook %d ID: ",i+1); scanf("%d",&books[i].id);
        printf("Title: "); scanf("%49[^\n]",books[i].title);
        printf("Author: "); scanf("%49[^\n]",books[i].author);
        printf("Year: "); scanf("%d",&books[i].year);
        books[i].available=1;
    }

    printf("\nBOOK LIST\n");
    for(i=0;i<n;i++)
        printf("%d | %s | %s | %d | %s\n",
            books[i].id,books[i].title,books[i].author,books[i].year,
            books[i].available?"Available":"Issued");
    return 0;
}
```

6. Sample Input

Number of books: 2
 101 Data Structures Lipschutz 2024
 102 C Programming Ritchie 2015

7. Sample Output

101 | Data Structures | Lipschutz | 2024 | Available
 102 | C Programming | Ritchie | 2015 | Available

8. Complexity

Time: $O(n)$ for adding/traversing n books. Direct index access: $O(1)$. Space: $O(n)$.

9. Practice Exercises

- Add category and price fields.
- Count available books.
- Display books published after 2020.

10. Related Viva Questions

- What is an array of structures?
- How do you access the third book?
- What is the complexity of traversal?
- Why is array indexing $O(1)$?